

Scientists & Inventions

Notes

Albert Einstein (1879-1955) :

- German-born physicist who developed the special and general theories of relativity and won the **Noble Prize for Physics** in 1921 for his explanation of the **photoelectric effect**.
- Einstein is generally considered the most influential physicist of the 20th Century.

Einstein miracle year 1905 –

- While working at the patent office, he did some of the most creative works of his life.
- **First Paper** – He applied the quantum theory (Developed by German Physicist Max Planck) to light in order to explain the phenomenon known as the photoelectric effect, by which a material will emit electrically charged particles when hit by light.
- **Second Paper** – This paper of Einstein contained Einstein's experimental proof of the existence of atoms, which he got by analyzing the phenomenon of Brownian Motion, in which tiny particles were suspended in water.
- **Third Paper** – In the third and most famous article titled “On the Electrodynamics of Moving Bodies” Einstein confronted the apparent contradiction between two principal theories of physics : **Issac Newton** concepts of absolute space and time and **James Clark Maxwell's** idea that the speed of light is a constant. To do this he introduced his **Special Theory of Relativity** which held that the laws of physics are the same even for objects moving in different inertial frames (i.e. at constant speeds relative to each other) and that the speed of light is constant in all inertial frames.
- A Fourth paper concerned the fundamental relationship between mass and energy. Einstein famous equation $E = mc^2$ (where ‘c’ is the constant speed of light) expressed this relationship.
- In 1915, Einstein published the **General Theory of Relativity** which explains the law of gravitation and its relation to the forces of nature.

C.V. Raman (1888-1970) :

- Chandrashekhara Venkata Raman was born at Thiruvanaikaval in Tiruchirappalli in Southern India on

Nov. 7, 1888. His father was a Lecturer in Mathematics and Physics so that from childhood he was immersed in an academic atmosphere. He entered Presidency College Madras in 1902 and in 1904 he completed his bachelor's degree winning the first place and the gold medal in Physics and in 1907 he gained his Master degree, obtaining the highest distinctions.

- His earliest researches in optics and acoustics-the two fields of investigation to which he has dedicated his entire career were carried out while he was a student.
- Since at that time a scientific career did not appear to present the best possibilities, Raman joined the Indian Finance Department in 1907, though the duties of his office took most of his time, Raman found opportunities for carrying on experimental research in the laboratory of the Indian Association for the Cultivation of Science at Calcutta (now Kolkata) of which he became Honorary Secretary in 1909.
- In 1917 he was offered the newly endowed Palit Chair of Physics at Calcutta University and decided to accept it. After 15 years at Calcutta he became Professor at the Indian Institute of Science at Bangalore.
- He was elected a Fellow of the **Royal Society** early in his career (1924) and was Knighted in 1929. In 1930 he was awarded **Nobel Prize** for his work **Raman-Effect**.

Satyendra Nath Bose (1894-1974) :

- Satyendra Nath Bose [Born Jan. 1, 1894, Calcutta (now Kolkata) – died Feb. 4, 1974], was great Indian Mathematician and Physicist voted for his collaboration with Albert Einstein in developing a theory regarding the gaslike qualities of Electromagnetic radiation (Bose - Einstein Statistics).
- Bose, a graduate of the University of Calcutta taught at the University of Dhaka (1921-1945) and then at Calcutta (1945-1956). Bose's numerous scientific papers published contributed to statistical mechanics and thermoluminescence and unified field theory. Bose's Planck's Law and the Hypothesis of Light Quanta (1924) led Einstein to seek him out for a collaboration.
- A Fellow of Royal Society, he was awarded India's second-highest civilian award, the Padma Vibhushan in 1954 by the Govt. of India.
- The class of particles that obey Bose-Einstein statistics- **bosons** was named after Bose by Paul Dirac.

Scientist	Main contribution / Invention	Native land
Archimedes	Principle of Buoyancy, Principle of Lever	Greek
Galileo Galilei	Telescope, Principle of inertia	Italy
Christian Huygens	Wave theory of light, Pendulum Clock, Discovery of Saturn's moon Titan	Holland
Issac Newton	Laws of motion, Law of universal gravitation, Reflecting telescope	England
Michael Faraday	Principles of electromagnetic induction	England
James Clerk Maxwell	Electromagnetic Principle, Light - An electromagnetic wave	England
Heinrich Rudolf Hertz	Electromagnetic Waves, Photoelectric effect	Germany
Jagdish Chandra Bose	Investigation of Radio and Microwave optics	India
W. C. Rontgen	X-Ray (Invention)	Germany
J. J. Thomson	Discovery of Electrons	England
Marie Sklodowska Curie	Discovery of Radium & Polonium, Study of Natural Radioactivity (Only person to win Nobel Prize in both Physics & Chemistry)	Polland
Albert Einstein	Theory of relativity, Photoelectric effect explanation	Germany
Victor Francis Hess	Discovery of Cosmic rays	Austria
Robert Andrews Millikan	Measurement of elementary electric charge	America
Ernest Rutherford	Discovery of Atomic nucleus, Proton, α , β rays, Set forth the laws of radioactive decay & identified α -particles as helium nuclei	New Zealand
Niels Bohr	Quantum mechanical model of Hydrogen atoms	Denmark
Chandrashekhara Venkata Raman	Discovery of Raman Effect (Inelastic scattering of light by molecules)	India
Louis Victor de Broglie	Wave Properties of matter	France
Meghnad Saha	Saha ionization equation	India
Satyendra Nath Bose	Boson - a class of particle named after him, Quantum Statistics	India
Wolfgang Ernst Pauli	Exclusion Principle	Austria
Enrico Fermi	Called as the architect of nuclear age and architect of the atomic bomb, Controlled nuclear fission	Italy
Werner Heisenberg	Quantum Mechanics, Uncertainty Principle	Germany
Paul Dirac	Quantum Mechanics, Quantum electrodynamics	England
Edwin Hubble	A pioneer in the field of extra galaxy astronomy, Expansion of Universe	America
Ernest Orlando Lawrence	Invented Cyclotron - a particle accelerator; Nowadays it is widely used in artificial transmutation	America
James Chadwick	Discovery of Neutron	England
Hideki Yukawa	Theory of nuclear forces, Prediction of the pi meson	Japan
Homi Jahangir Bhabha	An Indian Nuclear Physicist, Father of Indian Nuclear Programme, Cosmic Radiations	India
Lev Davidovich Landau	Quantum theory of condensed matter & liquid Helium	Russia
S. Chandra Shekhar	Chandra Shekhar limit and structure & development of stars	India
John Bardeen	Transistor, Theory of Superconductivity (He is the only person to be awarded the Noble Prize in Physics twice-first in 1956 for the invention of the transistor and again in 1972 for a fundamental theory of Conventional superconductivity known as the BCS-Bardeen-Cooper-Schrieffer Theory)	America
C.H. Townes	MASER and LASER	America
Abdus Salam	Electroweak unification theory	Pakistan

Technology	Scientific Principle
Steam Engine	Law of thermodynamics
Nuclear Reactor	Controlled nuclear fission
Radio and Television	Production, communication and detection of electromagnetic waves
Computer	Numerical logic
Production of high magnetic fields	Superconductivity
LASER	Optical amplification based on the stimulated emission of electromagnetic radiation
Rocket Propulsion	Newton's law of motion
Generator	Principles of electromagnetic induction of Faraday
Hydroelectric Power	Transformation of gravitational potential energy into electrical energy
Aeroplane	Bernoulli's principle in fluid dynamics
Particle accelerator	Velocity of charged particles in electromagnetic waves
SONAR	Reflection of Ultrasonic waves
Optical fibres	Total internal reflection of light
Antireflecting coating	Thin film optical interference
Electron microscope	Wave nature of electrons
Photoelectric cell	Photoelectric effect
Tokamak Fusion Test Reactor	Fusion reactions in the plasma
Giant Meterwave Radio Telescope (GMRT)	Detection of cosmic radio waves
Bose-Einstein condensate	Coding a gas of extremely low density about one hundredth to thousandth the density of normal air to ultra-low temperature

Question Bank

1. The machine-Gun was invented by –

- (a) J.L. Beared (b) G. Brousa
(c) Karl Benz (d) James Puckle

U.P.P.C.S. (Mains) 2014

Ans. (d)

In 1718 an Englishman named James Puckle (1667-1724) invented defence gun what proved to be one of the keys steps in the evolution of mechanical machine guns. That year Puckle, a London solicitor patented what he called the Defence Gun. There is some argument among historians about whether this was the first patent of a manually operated revolving-type machine gun, but the importance of Puckle's gun is that it certainly represents the most refined design to be found till that time.

2. **Who invented thermoscope, an early form of thermometer?**

- (a) Sir Christopher Wren (b) Charles F. Richter
(c) Galileo (d) Beno Gutenberg

M.P.P.C.S. (Pre) 2010

Ans. (c)

Thermometer or thermoscope was discovered by famous Italian physicist Galileo in 1609.

3. **Telescope was invented by :**

- (a) Galileo (b) Gutenberg
(c) Edison (d) Graham Bell

U.P.P.C.S. (Pre) 1994

Ans. (a)

Telescope was invented by Galileo in 1609.

4. **The microscope is used for the study of?**

- (a) Distant objects (b) Nearby objects
(c) Small objects (d) Micro and nearby objects

M.P.P.C.S. (Pre) 2015

Ans. (d)

A microscope is an equipment that helps in watching micro and nearby objects which cannot be seen by naked eyes.

5. **Match the following :**

- A. Typewriter** **1. Davy**
B. X-ray **2. Sholes**
C. Radio **3. Rontgen**
D. Safety Lamp **4. Marconi**

Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 2 | 3 | 4 | 1 |
| (b) | 1 | 2 | 3 | 4 |
| (c) | 3 | 4 | 1 | 2 |
| (d) | 4 | 1 | 2 | 3 |

M.P.P.C.S. (Pre) 1999

Ans. (a)

The typewriter was invented by Sholes in 1867. X-ray was invented by Wilhelm Rontgen in 1895, Radio was invented by Guglielmo Marconi in 1901. Safety lamp was invented by Humphry Davy in 1815.

6. **Wilhelm Roentgen invented –**

- (a) Radio (b) X-Ray Machine
(c) Electric Bulb (d) Electric Motor

U.P.P.C.S. (Pre) 2005

Ans. (b)

See the explanation of above question.

7. **Who among the following scientists shared the Nobel Prize in physics with his son ?**

- (a) Max Plank (b) Albert Einstein
(c) William Henry Bragg (d) Enrico Fermi

I.A.S. (Pre) 2008

Ans. (c)

British physicist and chemist Sir William Henry Bragg shared Nobel Prize in physics with his son William Lawrence Bragg in 1915. With the help of his son, he developed X-ray spectrometer for the analysis of crystal structure by mean of X-rays.

8. **Match the following**

- | | |
|---------------------------------|--------------------------|
| A. John Guttenberg | 1. Telephone |
| B. W.C. Rontgen | 2. Printing Press |
| C. Michael Faraday | 3. X-Ray |
| D. Alexander Graham Bell | 4. Dynamo |

Code :

- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 1 | 2 | 3 | 4 |
| (b) | 2 | 3 | 1 | 4 |
| (c) | 3 | 4 | 1 | 2 |
| (d) | 2 | 3 | 4 | 1 |
| (e) | 2 | 4 | 3 | 1 |

Chhattisgarh P.C.S (Pre) 2013

Ans. (d)

The correctly matched list is given below :

John Guttenberg	–	Printing Press
W.C. Rontgen	–	X-ray
Michael Faraday	–	Dynamo
Alexander Graham Bell	–	Telephone

9. **Who invented Steam Engine?**

- (a) Robert Watson (b) James Watt
(c) William Harvey (d) None of the above

Uttarakhand U.D.A./L.D.A. (Pre) 2007

Ans. (b)

The creation of steam engine began in late 17th century. In 1698, Thomas Savery an English inventor, patented a device that allowed the use of steam to pump water. The steam engine that he created was used to remove water from mines. The design was improved upon by English engineer Thomas

Newcomen in 1712. Later, further and more significant improvements to the steam engine were made by Scottish engineer and inventor James Watt in 1763. So due to this reason he is regarded as the inventor of the steam engine.

10. **'Telephone' was invented by –**

- (a) Graham Bell (b) Hoffman
(c) Wilson (d) George Stephenson

M.P.P.C.S. (Pre) 1995

Ans. (a)

Telephone was invented by Alexander Graham Bell in 1876.

11. **Who discovered electric bulb?**

- (a) Thomas Edison (b) Alexander Graham Bell
(c) William Cooke (d) Terry Addison

M.P.P.C.S. (Pre) 2010

Ans. (a)

American scientist Thomas Alva Edison is the inventor of the electric bulb.

12. **Gas engine was invented by :**

- (a) Diesel (b) Davy
(c) Daimler (d) Charles

U.P.P.C.S. (Pre) 1995

Ans. (c)

Diesel engine was developed by Rudolph Diesel in 1895. Gas engine was developed by Gottlieb Daimler in 1885.

13. **Who was the inventor of the fountain pen?**

- (a) Waterman (b) Parker
(c) Chelpark (d) Shaeffers

M.P.P.C.S. (Pre) 2005

Ans. (a)

Lewis Edson Waterman invented fountain pen in 1883 in New York. He patented the first practical fountain pen in 1884.

14. **Who was the inventor of Radar –**

- (a) Robert Watson (b) Fleming
(c) Bush Wall (d) Austin

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (a)

Scottish physicist Robert Alexander Watson was a pioneer and significant contributor to the development of Radar (Radio Detection and Ranging). He produced the first radar system in 1935. This system was greatly used by Royal Air Force of Britain in World War II.

15. **Alfred Nobel invented –**

- (a) Microphone (b) Typewriter
(c) Dynamite (d) Gramophone

M.P.P.C.S. (Pre) 1995

Ans. (c)

Swedish scientist Alfred Nobel invented dynamite. The famous Nobel Foundation was established on the basis of his will in 1900.

16. Who discovered Television?

- (a) Edison (b) Galileo
(c) Baird (d) Franklin

M.P.P.C.S. (Pre) 1995

Ans. (c)

Television was invented by J.L. Baird in 1926 in Britain. Galileo invented the Telescope.

17. With what invention is the name of J.L. Baird associated with?

- (a) Barometer (b) Helicopter
(c) Television (d) Telephone

M.P.P.C.S. (Pre) 1998

Ans. (c)

See the explanation of above question.

18. Who among the following invented Laser ?

- (a) Theodore Maiman (b) Denis Papin
(c) William Morton (d) Francis Crick

Uttarakhand Lower Sub. (Pre) 2010

I.A.S. (Pre) 2005

Ans. (a)

Theodore Maiman was the inventor of Laser (Light Amplification by the Stimulated Emission of Radiation). On 16 May, 1960 he successfully demonstrated the laser in action from a ruby crystal in his laboratory at Hughes in Malibu.

19. Which one of the following pairs is NOT correctly matched?

- (a) James Watt : Steam Engine
(b) A.G. Bell : Telephone
(c) J.L. Baird : Television
(d) J. Perkins : Penicillin

U.P.P.C.S (Pre) 2011

Ans. (d)

Penicillin was discovered by Scottish scientist Alexander Fleming. Rest pairs are correctly matched.

20. Who is the discoverer of Penicillin?

- (a) Alexander Graham Bell
(b) Alexander Fleming
(c) Christopher Macdonald
(d) Albert Einstein

M.P.P.C.S. (Pre) 2010

Ans. (b)

See the explanation of above question.

21. Alexander Fleming discovered-

- (a) Penicillin (b) X-ray
(c) Streptomycin (d) Telephone

M.P.P.C.S. (Pre) 2015

Ans. (a)

See the explanation of above question.

22. What did Sir Alexander Fleming discover ?

- (a) Penicillin (b) Blood Circulation
(c) Oxygen (d) Antiseptic Medicine

M.P.C.C.S. (Pre) 1998

Ans. (a)

See the explanation of above question.

23. What did Fleming discover ?

- (a) Radium (b) Penicillin
(c) Vaccine (d) Anesthetic

Uttarakhand Lower Sub. (Pre) 2010

Ans. (b)

See the explanation of above question.

24. Match the following in the correct order –

- | | |
|----------------------|-----------------------------|
| A. Radium | 1. Alexander Fleming |
| B. Penicillin | 2. Madam Curie |
| C. X-Ray | 3. Edward Jenner |
| D. Measles | 4. W.C. Roentgen |

Code :

- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 2 | 1 | 4 | 3 |
| (b) | 2 | 3 | 4 | 1 |
| (c) | 3 | 4 | 1 | 2 |
| (d) | 4 | 1 | 2 | 3 |

U.P. P.C.S. (Pre) 1992

Ans. (a)

Radium was discovered by Pierre Curie and Madam Curie in 1898. Penicillin was discovered by Alexander-Fleming in 1928. X-Ray was discovered by W.C. Roentgen in 1895 and Edward Jenner discovered the vaccine for measles (Small Pox) in 1796.

25. Who among the following is the inventor of 'X' rays?

- (a) Rutherford (b) Roentgen
(c) Maxwell (d) Torricelli

Uttarakhand U.D.A./L.D.A. (Pre) 2003

Ans. (b)

See the explanation of above question.

26. Arrange the following in chronological order as per their discovery and select correct answer from the code given below:

- A. Transistor**
B. Diode

C. SMD (Surface Mounted Device)

D. Integrated Circuit (IC)

Code :

(a) A, B, C, D

(b) B, A, C, D

(c) B, A, D, C

(d) A, C, B, D

U.P. R.O./A.R.O. (Mains) 2016

Ans. (c)

The transistor was invented in 1947 by American physicists John Bardeen and Walter Brattain while working under William Shockley at Bell Labs, USA. The three shared the 1956 Nobel Prize in Physics for their achievement. John Ambrose Fleming invented the first working diode in 1904. Surface Mounted Device (SMD) was developed by American technology company IBM in the 1960s. Jack Kilby and Robert Noyce developed first working integrated circuits (ICs) in 1958 and 1959, respectively. Hence, the correct chronological order is given in option (c).

27. Match List-I with List-II and select the correct answer using the codes given below :

List-I

A. Alfred Nobel

B. Alexander Fleming

C. Charles Darwin

D. Madam Curie

List-II

1. Theory of Evolution

2. Dynamite

3. Penicillin

4. Isolation of radium

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	2	3	1	4
(c)	3	2	1	4
(d)	1	4	3	2

U.P.P.C.S. (Mains) 2010

Ans. (b)

Alfred Bernhard Nobel was a Swedish chemist engineer, innovator. He was the inventor of dynamite. Alexander Fleming was a Scottish biologist. His best-known discoveries are the enzyme lysozyme in 1923 and the antibiotic 'penicillin' from the fungus *Penicillium notatum* in 1928, for which he got the Nobel Prize in Physiology or Medicine in 1945 with Howard Florey and Ernst Boris Chain. Charles Robert Darwin was an English naturalist and geologist, best known for his contributions to evolutionary theory. Madam Curie was a Polish and naturalized-French physicist and chemist, who conducted pioneering research on radioactivity.

28. The principle of 'Black hole' was enunciated by :

(a) C.V Raman

(b) H.J. Bhabha

(c) S. Chandrasekhar

(d) H. Khurana

U.P.P.C.S (Pre) 2011

Ans. (c)

Indian American astrophysicist Subrahmanyam Chandrasekhar was best known for rendering the theory of 'Chandra Sekhar Limit'. For this, he used Albert Einstein's special theory of relativity and the principle of quantum physics. Chandrasekhar showed that it is impossible for a white dwarf star to be stable if its mass is greater than 1.44 times that of the sun. If such a star does not completely exhaust its thermonuclear fuel, then this limiting mass may be slightly larger. A star that ends its nuclear-burning lifetime with a mass greater than the Chandrasekhar limit must be either a Neutron star or a black hole. For this contribution in astrophysics he was awarded the 1983 Nobel Prize in Physics with William A. Fowler.

29. Scientist S. Chandrasekhar got Nobel Prize for which field?

(a) Astrophysics

(b) Micro-electronics in Satellites

(c) Super conductivity at low temperatures

(d) Studies of Core of the sun.

M.P.P.C.S. (Pre) 1996

Ans. (a)

See the explanation of above question.

30. Who of the following scientists proved that the stars with mass less than 1.44 times the mass of the sun end up as White Dwarfs when they die?

(a) Edwin Hubble

(b) S. Chandra Shekhar

(c) Stephen Hawking

(d) Steven Weinberg

I.A.S. (Pre) 2009

Ans. (b)

S. Chandrasekhar proved that the stars with mass less than 1.44 times the mass of the sun ends up as white dwarfs when they die. He showed that the mass of a white dwarf could not exceed 1.44 times that of the sun.

31. The photoelectric effect was discovered by :

(a) Hertz

(b) Einstein

(c) Plank

(d) Bohr

U.P.P.C.S. (Mains) 2003

Ans. (a)

Photoelectric effect is a phenomenon in which electrically charged particles are released from or within a material when it absorbs electromagnetic radiation. The effect is often defined as the emission of electron from a metal plate when light falls on it. This effect was discovered in 1887 by the German Physicist Heinrich Rudolf Hertz. This is also known as Hertz effect.

32. Scientist Albert Einstein is famous for –
 (a) The interpretation of atomic structure of hydrogen
 (b) The photoelectric effect
 (c) Planting first Nuclear Reactor
 (d) The prediction of the existence of neutrons

38th B.P.S.C. (Pre) 1992

Ans. (b)

Scientist Albert Einstein is famous for giving the simple and factual explanation of Photoelectric effect on the basis of Planck's quantum theory. For this, he was awarded Nobel Prize in Physics in 1921.

33. Einstein was awarded Nobel Prize for –
 (a) Theory of relativity (b) Brownian motion
 (c) Photoelectric effect (d) Specific heat of solids

U.P.P.C.S. (Pre) 2015

Ans. (c)

See the explanation of above question.

34. Einstein was honoured with Nobel Prize for :
 (a) Photoelectric Effect
 (b) Theory of Specific Heat
 (c) Special Theory of Relativity
 (d) Bose-Einstein Statistics

U.P. P.C.S. (Pre) 2016

U.P. Lower Sub. (Pre) 2015

Ans. (a)

See the explanation of above question.

35. Einstein was awarded Nobel Prize for which of his theories?
 (a) Theory of relativity
 (b) Theory of photoelectricity
 (c) Theory of specific heats
 (d) Theory of Brownian motion

U.P. P.C.S. (Pre) 2018

Ans. (b)

See the explanation of above question.

36. Einstein got the Nobel Prize for :
 (a) relativity
 (b) Bose-Einstein condensation
 (c) Mass-energy equivalence
 (d) photoelectric effect
 (e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (d)

See the explanation of the above question.

37. The 'theory of relativity' is presented by which scientist?
 (a) Albert Einstein (b) Isaac Newton

- (c) Stephen Hawking (d) Marie Curie
 (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (a)

The 'theory of relativity' usually encompasses two interrelated theories by Albert Einstein : special relativity and general relativity, proposed and published in 1905 and 1915, respectively. Special relativity applies to all physical phenomena in the absence of gravity. Special relativity argued that space and time are inextricably connected, but that theory didn't acknowledge the existence of gravity. General relativity explains the law of gravitation and its relation to the forces of nature. General relativity is Einstein's understanding of how gravity affects the fabric of space-time. It applies to the cosmological and astrophysical realm, including astronomy.

38. In Einstein's equation, $E = mc^2$, c denotes –

- (a) Velocity of sound (b) Speed of light
 (c) Wavelength (d) A constant

Uttarakhand P.C.S. (Pre) 2007

Ans. (b)

Einstein equation shows that mass is equivalent to energy, whether it might be in any form. According to Einstein's formula $E = mc^2$
 where, E = Energy
 m = mass
 c = speed of light

39. Albert Einstein was proficient in playing which musical instrument ?

- (a) Guitar (b) Flute
 (c) Violin (d) Sitar

Uttarakhand P.C.S. (Pre) 2007

Ans. (c)

Albert Einstein was proficient in playing violin and piano.

40. The fourth dimension in Physics was introduced by :

- (a) Newton (b) Einstein
 (c) Galileo (d) Neil Bohr

Uttarakhand P.C.S. (Pre) 2007

Ans. (b)

The fourth dimension can refer to time as another dimension, along with length, width, and depth. This idea of time as a fourth dimension is usually attributed to the 'Theory of Special Relativity' proposed in 1905 by the German-born physicist Albert Einstein.

41. Sir C.V. Raman received Nobel Prize for physics in the year :

- (a) 1928 (b) 1930

(c) 1932

(d) 1950

42nd B.P.S.C. (Pre) 1997

Ans. (b)

In 1930, C.V. Raman was the first non-white, Asian and Indian to receive the Nobel Prize in Physics for the work on scattering of light and discovery of Raman Effect. Every year 28th February is celebrated as National Science Day. It is because on the same day in 1928 he discovered 'Raman Effect'.

42. An eminent Indian physicist Dr. C. V. Raman was given Nobel Prize for his work in the year :

(a) 1925

(b) 1930

(c) 1935

(d) None of these

Uttarakhand U.D.A./L.D.A. (Pre) 2007

Ans. (b)

See the explanation of above question.

43. The National Science Day is observed on February 28 every year to commemorate :

(a) Raman-effect day

(b) The launch of the first artificial satellite Aryabhata

(c) The successful launch of ASLV

(d) Homi J. Bhabha's birthday

U.P. P.C.S. (Pre) 2003

U.P. U.D.A./L.D.A. (Pre) 2002

U.P. P.C.S. (Pre) 1990

U.P. P.C.S. (Pre) 1993

Ans. (a)

See the explanation of above question.

44. The National Science Day is observed on :

(a) January, 28

(b) February, 28

(c) March, 28

(d) April, 28

Uttarakhand P.C.S. (Mains) 2006

U.P. Lower Sub. (Spl.) (Pre) 2002, 2003, 2004

Ans. (b)

See the explanation of above question.

45. India celebrates the National Science Day on :

(a) June, 5

(b) December, 22

(c) February, 28

(d) January, 30

U.P. P.C.S. (Mains) 2016

Ans. (c)

See the explanation of above question.

46. Which scientist birthday is celebrated as the 'National Science Day'?

(a) Ramanujan

(b) Prof. S. Chandrashekhar

(c) C.V. Raman

(d) Dr. H.J. Bhabha

M.P. P.C.S. (Pre) 2004

Ans. (*)

Famous scientist C.V. Raman was born on November 7, 1888 and 'National Science Day' is celebrated on 28th February, so the question is incorrect. National Science Day is the day when C.V. Raman discovered the Raman Effect.

47. Whose birth centenary was celebrated in the year 1988?

(a) Charles Darwin

(b) Einstein's Theory of Relativity

(c) Mathematician Ramanujan

(d) C.V. Raman

U.P.P.C.S. (Pre) 1991

Ans. (d)

Charles Darwin (February 12, 1809 - April 19, 1882)
Ramanujan (December 22, 1887 - April 26, 1920)
Einstein proposed the Special Theory of Relativity in 1905 and General Theory of Relativity in 1915.
C.V. Raman (November 7, 1888 - November 21, 1970)
Thus, it is clear that in the year 1988 birth centenary of C.V. Raman was celebrated.

48. A recent movie titled *The Man Who Knew Infinity* is based on the biography of

(a) S. Ramanujan

(b) S. Chandrashekhar

(c) S. N. Bose

(d) C.V. Raman

I.A.S. (Pre) 2016

Ans. (a)

The Man Who Knew Infinity is a 2015 British biographical drama film based on the 1991 book of the same name by Robert Kanigel. The film stars Dev Patel as Srinivasa Ramanujan, a real life mathematician who after growing up poor in Madras earns admittance to Cambridge University, where he becomes a pioneer in Mathematical theories.

49. Who among the following scientists propounded the New Theory of Relativity?

(a) J. V. Narlikar

(b) M. M. Krishnan

(c) S. Chandrashekhar

(d) B. D. Nag Choudhary

U.P. P.C.S. (Mains) 2016

Ans. (a)

J.V. Narlikar propounded the 'new theory of relativity'. In collaboration with a senior scientist Fred Hoyle, he propounded this theory by amending the Einstein's Theory of Relativity.

50. According to J.V. Narlikar which of the following has so far NOT been considered as a science?

(a) Astrology

(b) Astronomy

(c) Cosmology

(d) Nanotechnology

U.P. P.C.S. (Pre) 2018

Ans. (a)

According to Indian astrophysicist Jayant Vishnu (J.V.) Narlikar, astrology is not a science.

51. Who is called the Father of Atomic Energy in India ?

- (a) Prof. Satish Dhawan
- (b) Homi J. Bhabha
- (c) Dr.K.S. Krishna
- (d) None of the above

Uttarakhand U.D.A./L.D.A. (Pre) 2007

Ans. (b)

Homi Jehangir Bhabha was an Indian nuclear physicist. He played significant role in development of nuclear energy programme in India. He is known as the father of Indian nuclear programme.

52. Homi Bhabha Award is given for special contribution in the field of –

- (a) Theoretical Physics
- (b) Nuclear Energy
- (c) Laser Physics
- (d) Space Research

U.P.P.C.S.(Pre) 2012

Ans. (b)

Homi Jehangir Bhabha is known as the father of nuclear energy programme in India. So in his memory, the 'Homi Bhabha Award' is given to the scientists, especially for their contribution in the field of nuclear energy.

53. Stephen Hawking is a –

- (a) Pianist
- (b) Guitarist
- (c) Scientist
- (d) American politician

M.P.P.C.S. (Pre) 2010

Ans. (c)

Stephen William Hawking (8 January 1942 - 14 March 2018) was a world famous British theoretical physicist, cosmologist, and author who was director of research at the Centre for Theoretical Cosmology at the University of Cambridge at the time of his death. He is known for his work regarding black holes and for authoring several popular science books. He suffered from Amyotrophic Lateral Sclerosis (ALS) commonly known as motor neuron disease.

54. Which one of the following scientists has carried out researches both in the field of biology and physics?

- (a) Jagdish Chandra Bose
- (b) Har Govind Khurana
- (c) C. V. Raman
- (d) Homi J. Bhabha

U.P. Lower Sub. (Pre) 2008

Ans. (a)

Jagdish Chandra Bose was an eminent Indian scientist who carried out researches both in the field of biology and physics. He made a significant contribution in the field of botany along with radio and microwave signals. He also invented the Crescograph for measuring the growth in plants using a series of clockwise gears.

55. Match the names of outstanding Indian scientists given in List-I with an area of their specialized work given in List-II and select the correct answer by using the codes given below the lists :

List-I

- A. Dr. Raja Ramanna
- B. Dr. M.S. Swaminathan
- C. Prof. U.R. Rao

List-II

- 1. Plant chemistry
- 2. Nuclear physics
- 3. Thermodynamics and astrophysics
- 4. Space research
- 5. Agricultural science

D. Prof. Meghnad Saha

Code :

	A	B	C	D
(a)	3	5	2	1
(b)	2	1	4	3
(c)	2	5	4	3
(d)	2	1	4	2

I.A.S. (Pre) 1998

Ans. (c)

Dr. Raja Ramanna was an Indian physicist who is best known for his role in India's nuclear programme. Dr. M.S. Swaminathan is a famous Indian agricultural scientist. He is known as the 'Father of Green Revolution in India'. U.R. Rao is well-known space scientist and former Chairman of the Indian Space Research Organisation. Prof. Meghnad Saha was an Indian astrophysicist best known for his development of the Saha ionization equation.

56. The principle of Thermal Ionisation is an outstanding contribution of :

- (a) H.J. Bhabha
- (b) M.N. Saha
- (c) C.V. Raman
- (d) J.C. Bose

U.P.P.C.S. (Mains) 2014

Ans. (b)

The theory of Thermal Ionization of gases was first given by Meghnad Saha in a series of papers (1920 a, 1920 b, 1921) and widely applied by him and others to explain the spectrum of the Sun and furnish a satisfactory physical theory of stellar spectra and their classification.

57. Who is associated with the development of India's Atomic Bomb :

- (a) A.P.J. Abdul Kalam
- (b) Homi Bhabha
- (c) Raja Ramanna
- (d) Kastoorirangan

U.P. Lower Sub. (Pre) 2004

U.P.P.C.S. (Pre) 2002

Ans. (c)

Indian physicist Dr. Raja Ramanna contributed a lot to the development of India's Atomic Bomb. Under his supervision, the first nuclear test (codename 'Smiling Buddha') was carried out in 1974. He also served as Director General of DRDO and as Scientific Advisor to the Defence Minister of India during 1978-81.